

CURRICULUM OUTLINE

THE BLUEPRINT TO SUCCESS

Our expert-led curriculum will prepare you for the toughest challenges that you may face in your journey to becoming a skilled programmer.

COURSE OUTLINE FOR BEGINNERS

Segment	Module Type	Module name	Duration (months)
Programming Fundamentals	Core	Programming Fundamentals - Thinking with AI: Logics, Loops & Control Flow	1
	Core	Programming Fundamentals - Code Smarter with AI: Function & Arrays	1
	Core	Intermediate DSA : AI Assisted Problem Solving	1
Agentic AI	Core	AI & Agents: From Talking to AI to Building One	1
Advanced DSA	Core	Advanced DSA: Foundations:Core Techniques & Optimization	1
	Core	Advanced DSA: Linear & Non-Linear Structures	1
	Core	Advanced DSA: Backtracking & Advanced Trees	1
	Core	Advanced DSA: DP, Heaps & Graphs — Crack It with AI Edge	1
Databases & SQL	Core	Databases & SQL	1
Backend	Specialisation	Foundations of Object-Oriented Design & Scalable Systems	1
	Specialisation	Design Principles & Patterns for Extensible Systems	1
	Specialisation	Applied Design & Machine Coding	1
	Specialisation	AI-First Backend Capstone Project primer	1
	Elective	AI-First Backend Capstone Project Advance	1.5
	Elective	Build a Production-Ready AI-Enabled System	1
Fullstack	Specialisation	Web Platform Foundations & UI Systems	0.5
	Specialisation	JavaScript Runtime Systems, Browser APIs & Product Engineering Labs	1.5
	Specialisation	React and Frontend Product Architecture Fundamentals	1
	Specialisation	Backend Product Systems, APIs and Production Engineering	1
	Elective	Backend Product Systems - Deep Dive	0.5

	Elective	Foundations of AI-Assisted Full-Stack Development	0.5
	Elective	JavaScript Object System & Engineering Patterns	0.5
	Elective	React and Frontend Product Architecture Fundamentals Deep Dive	1
FDE	Specialisation	Foundations of the Forward Deployed Engineer Role & Production Python	7.5
	Specialisation	Backend Systems, Observability & Advanced Data Engineering	
	Specialisation	AI-First Frontends with TypeScript & React	
	Specialisation	Cloud, DevOps & Kubernetes for Production Delivery	
	Specialisation	LLM Engineering & Customer Consulting (RAG, Prompting & Guardrails)	
	Specialisation	Agentic Systems & Enterprise Integrations (Agents, MCP & HITL)	
	Specialisation	Secure, Multi-Tenant Application Delivery & Incident Response	
	Specialisation	FDE Capstone: End-to-End Customer Engagement (Discovery to Demo Day)	
	Elective	Containerization and Orchestration	1.5
	Elective	AWS : multi-region cloud deep dive- 1.5 months	1.5
	Elective	Storage, Networking & Reliability for FDEs- storage architecture, DR, sharding, CDN, HA, fault tolerance - 1 month	1
System Design	Elective	Distributed System Design & AI-Integrated Architectures	1.5
Interview Preparation		Mastering LastMile - Interview Simulation	1.5 weeks
Data Engineering		Data Engineering	2
Advanced DSA		DSA Edge: Advanced Patterns & Interview Techniques	1
		Product Management for Engineers	1
		Gen AI for Software Engineers	1.5

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** Only 1 Specialisation is allowed per learner. Recorded lectures for the other can be availed via Student Dashboard. At Least 1 Specialisation is mandatory for successful completion of the course.

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COURSE OUTLINE FOR INTERMEDIATE

Segment	Module Type	Module name	Duration (months)
Programming Fundamentals	Core	Intermediate DSA : AI Assisted Problem Solving	1
Agentic AI	Core	AI & Agents: From Talking to AI to Building One	1
Advanced DSA	Core	Advanced DSA: Foundations:Core Techniques & Optimization	1
	Core	Advanced DSA: Linear & Non-Linear Structures	1
	Core	Advanced DSA: Backtracking & Advanced Trees	1
	Core	Advanced DSA: DP, Heaps & Graphs — Crack It with AI Edge	1
Databases & SQL	Core	Databases & SQL	1
Backend	Specialisation	Foundations of Object-Oriented Design & Scalable Systems	1
	Specialisation	Design Principles & Patterns for Extensible Systems	1
	Specialisation	Applied Design & Machine Coding	1
	Specialisation	AI-First Backend Capstone Project primer	1
	Specialisation	AI-First Backend Capstone Project Advance	1.5
	Elective	Build a Production-Ready AI-Enabled System	1
Fullstack	Specialisation	Web Platform Foundations & UI Systems	0.5
	Specialisation	JavaScript Runtime Systems, Browser APIs	1.5
	Specialisation	React and Frontend Product Architecture Fundamentals fundamentals	1
	Specialisation	Backend Product Systems, APIs and Production Engineering	1
	Specialisation	Backend Product Systems, Deep Dive	0.5
	Elective	Foundations of AI-Assisted Full-Stack Development	0.5
	Elective	Frontend Systems & Machine Coding Mastery	0.5
	Elective	React and Frontend Product Architecture Fundamentals Deep Dive	1
FDE	Specialisation	Foundations of the Forward Deployed Engineer Role & Production Python	7.5
	Specialisation	Backend Systems, Observability & Advanced Data Engineering	
	Specialisation	AI-First Frontends with TypeScript & React	
	Specialisation	Cloud, DevOps & Kubernetes for Production Delivery	
	Specialisation	LLM Engineering & Customer Consulting (RAG, Prompting & Guardrails)	
	Specialisation	Agentic Systems & Enterprise Integrations (Agents, MCP & HITL)	

	Specialisation	Secure, Multi-Tenant Application Delivery & Incident Response	
	Specialisation	FDE Capstone: End-to-End Customer Engagement (Discovery to Demo Day)	
	Elective	Containerization and Orchestration	1.5
	Elective	AWS : multi-region cloud deep dive- 1.5 months	1.5
	Elective	Storage, Networking & Reliability for FDEs- storage architecture, DR, sharding, CDN, HA, fault tolerance - 1 month	1
System Design	Elective	Distributed System Design & AI-Integrated Architectures	1.5
Interview Preparation		Mastering LastMile - Interview Simulation	1.5 weeks
Data Engineering		Data Engineering	2
Advanced DSA		DSA Edge: Advanced Patterns & Interview Techniques	1
		Product Management for Engineers	1
		Gen AI for Software Engineers	1.5

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COURSE OUTLINE FOR ADVANCED

Segment	Module Type	Module name	Duration (months)
Agentic AI	Core	AI & Agents: From Talking to AI to Building One	1
Advanced DSA	Core	Advanced DSA: Foundations:Core Techniques & Optimization	1
	Core	Advanced DSA: Linear & Non-Linear Structures	1
	Core	Advanced DSA: Backtracking & Advanced Trees	1
	Core	Advanced DSA: DP, Heaps & Graphs — Crack It with AI Edge	1
Databases & SQL	Core	Databases & SQL	1
Backend	Specialisation	Foundations of Object-Oriented Design & Scalable Systems	1

		Specialisation	Design Principles & Patterns for Extensible Systems	1
		Specialisation	Applied Design & Machine Coding	1
		Specialisation	AI-First Backend Capstone Project primer	1
		Specialisation	AI-First Backend Capstone Project Advance	1.5
		Elective	Build a Production-Ready AI-Enabled System	1
Fullstack		Specialisation	Web Platform Foundations & UI Systems	0.5
		Specialisation	JavaScript Runtime Systems, Browser APIs	1.5
		Specialisation	React and Frontend Product Architecture Fundamentals fundamentals	1
		Specialisation	Backend Product Systems, APIs and Production Engineering	1
		Specialisation	Backend Product Systems, Deep Dive	0.5
		Elective	Foundations of AI-Assisted Full-Stack Development	0.5
		Elective	Frontend Systems & Machine Coding Mastery	0.5
		Elective	React and Frontend Product Architecture Fundamentals Deep Dive	1
FDE		Specialisation	Foundations of the Forward Deployed Engineer Role & Production Python	7.5
		Specialisation	Backend Systems, Observability & Advanced Data Engineering	
		Specialisation	AI-First Frontends with TypeScript & React	
		Specialisation	Cloud, DevOps & Kubernetes for Production Delivery	
		Specialisation	LLM Engineering & Customer Consulting (RAG, Prompting & Guardrails)	
		Specialisation	Agentic Systems & Enterprise Integrations (Agents, MCP & HITL)	
		Specialisation	Secure, Multi-Tenant Application Delivery & Incident Response	
		Specialisation	FDE Capstone: End-to-End Customer Engagement (Discovery to Demo Day)	
		Elective	Containerization and Orchestration	1.5
		Elective	AWS : multi-region cloud deep dive- 1.5 months	1.5
		Elective	Storage, Networking & Reliability for FDEs- storage architecture, DR, sharding, CDN, HA, fault tolerance - 1 month	1
System Design		core	Distributed System Design & AI-Integrated Architectures	1.5
Interview Preparation		Elective	Mastering LastMile - Interview Simulation	1.5 weeks
Data Engineering			Data Engineering	2
Advanced DSA			DSA Edge: Advanced Patterns & Interview Techniques	1
			Product Management for Engineers	1
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CURRICULUM DEEP DIVE

Curriculum Deep Dive for Beginners

Programming Fundamentals ●

Programming Fundamentals - Thinking with AI: Logics, Loops & Control Flow

Programming Foundations & AI Awareness

Operators & Expression Evaluation using AI

Data Types 1

Data Types 2 + Reading Inputs

Operators

If-Else & AI Edge Case Thinking

Lab Session on If-Else

Loops & Iteration Analysis by AI

Lab Session on Loop

More problems on Loops & Patterns

Lab Session on Patterns

Dual Camera Proctoring

Data Types, If-Else & Loops & Contest discussion

Programming Fundamentals - Code Smarter with AI: Function & Arrays

Functions & Modularity: Designing Code with AI

Lab Session on Functions

Introduction to Strings

1D Array

Lab Session on 1D Array - 1

Lab Session on 1D Array - 2

2D Array - 1

Lab Session on 2D Arrays - 1

Lab Session on 2D Arrays - 2

Strings in Memory: Visualizing with AI

Maths Basics & Calculate Iterations

Revision: Language

Break

Contest 2: Functions & Arrays

Intermediate DSA : AI Assisted Problem Solving

Time Complexity

Introduction to Arrays & Basics of AI

Lab Session on TC, SC, Output & Debugging

Arrays: Prefix Sum & Carry Forward with AI-Assisted Problem Solving

Lab Session on Prefix Sum & Carry Forward

Arrays: Arrays techniques & smart prompting

Lab Session on Memory Management & Sorting Basics

Bit Manipulations Basics

Lab Session on 2D Matrices

Strings & Immutability visualisations through AI

Lab on Matrix & Strings using AI

Contest

Agentic AI ●

AI & Agents: From Talking to AI to Building One

GenAI World & LLM Landscape

Prompt Engineering Basics

Advanced Prompting & AI Safety Basics

LLM Built-in Power Tools

RAG

n8n Automation Workflow

What Is an AI Agent?

Building Agents with n8n

OpenClaw: Personal 24/7 AI Agent

Multi-Agent Systems

Debugging AI

Advanced DSA ●

Advanced DSA: Foundations:Core Techniques & Optimization

Time Complexity

Arrays Techniques

Arrays 1: One Dimensional

Arrays 2: Two Dimensional

Lab Session on Arrays

Bit Manipulation

Lab Session on Bit Manipulation

Recursion

Lab Session on Recursion

Maths: Modular Arithmetic & GCD

Hashing

Lab Session on Hashing

Sorting 1: Count Sort & Merge Sort

Sorting 2: Quick Sort & Comparator Problems

Sample Contest: Dual Camera Proctoring

Contest 1: Arrays, Bit Manipulation, Recursion, Math, Hashing & Sorting

Advanced DSA: Linear & Non-Linear Structures

Searching 1: Binary Search on Array

Searching 2: Binary Search on Answer

Lab Session on Searching

Classes, Objects & Linked List Introduction

Linked List: Basic Problems

Stacks

Lab Session on Stacks

Queues: Implementation & Problems

Trees 1: Structure & Traversal

Trees 2: BST

Lab Session on Binary Trees

Revision of DSA 1 & 2

Break

Contest 2: Searching, Linked List, Stacks, Queues & Trees

Advanced DSA: Backtracking & Advanced Trees

Maths: Combinatorics Basics & Prime Numbers

Lab Session on Prime Numbers & 2 Pointers

Lab Session on Maths & 2 Pointers

Backtracking

Lab Session on Backtracking

Linked List: Sorting and Problems

Linked List: Doubly Linked List & Detecting Loop

Trees 4: Morris Inorder Traversal + LCA

Lab Session on Binary Trees 2

Hashing 3: Internal Implementation & Problems

Contest 3: Math, Two Pointers, Backtracking, Linked List & Trees

Advanced DSA: DP, Heaps & Graphs — Crack It with AI Edge

Heaps: Introduction

Heap Sort & Greedy

Lab Session on Heaps & Greedy

Lab Session on Interview Problems 1

DP 1: One Dimensional

DP 2: Two Dimensional

DP 3: Knapsack

Lab Session on Applications of Knapsack

Graphs 1: Introduction, DFS & Cycle Detection

Graphs 2: BFS & MST

Graphs 3: Dijkstra Algo & Topological Sort

Lab Session on Interview Problems 2

Revision of DSA 3 & 4

Contest 4: Heaps, Greedy, DP & Graphs

Mandatory Skill Evaluation Test: DSA

Skill Evaluation Test Discussion + How to ace DSA Interview

DSA : Real World Project 1 (Social Network Analyzer)

DSA : Real world Project 2 (Invoice Merger)

Databases & SQL ●

Databases & SQL

Introduction to DBMS & Keys

CRUD Operations - 1

CRUD Operations - 2

Joins

Aggregate Queries

Joins and Aggregate Queries Working Session

Subqueries & Views

Indexing

Transactions

Window Functions

Schema Design - 1

Schema Design - 2

Databases & SQL Contest: SQL and Schema Design

Backend ●

Foundations of Object-Oriented Design & Scalable Systems

Intro to LLD for Scalable & AI-Ready Systems

OOP-1: Intro to OOP Lab

OOP-2: Access Modifiers and Constructors

OOP-3: Inheritance and Polymorphism

OOP-4: Interfaces, Abstract Classes

Concurrency-1: Introduction to Processes and Threads

Concurrency-2: Executors and Callables

Concurrency-3: Introduction to Synchronization

Concurrency-4: Synchronization with Semaphores

Java Advanced Concepts - 1 [Generics]

Java Advanced Concepts - 2 [Generics & Collections Lab]

Java Advanced Concepts - 3 [Streams and Lambdas]

Java Advanced Concepts - 4 [Exception Handling and Miscellaneous Topics]

Contest - 1: Java, OOP, and Concurrency

Design Principles & Patterns for Extensible Systems

SOLID Principles for Extensible Systems with AI integrations

Design Patterns: Introduction and Singleton

Design Patterns: Builder

Design Patterns: Prototype and Registry

Design Patterns: Leveraging Factory Pattern for Service Abstraction (APIs & AI Services)

Design Patterns: Adapter and Facade

Design Patterns: Decorator and Flyweight

Design Patterns: Behavioural Design Patterns

UML Diagrams

Contest - 2: Design Principles & Design Patterns

Applied Design & Machine Coding

How to Approach Design Problems

Design TicTacToe

Code TicTacToe 1

Code TicTacToe 2

Design Parking Lot

Code Parking Lot 1

Design BookMyShow

Code BookMyShow 1

Code BookMyShow 2

Code BookMyShow 3

Design Splitwise

Code Splitwise 1

Code Splitwise 2

Machine Coding

AI-First Backend Capstone Project Primer

Intro to Backend Systems & AI-Augmented Development

Version Control and Git

Learning Git Commands & AI Dev Tools

Intro to Spring Framework and Building APIs

MVC, Requests in Spring and Starting First Microservice (with AI Service Integration)

Calling 3rd Party APIs & LLM/AI Services

Intro to Spring Data JPA

JPA Queries and Repositories

Unit Testing and Good Practices (including AI-assisted testing)

Authentication vs Authorization, OAuth2, JWT

Implementing UserAuthService

Deploying Backend & AI-Enabled Applications to AWS

AI-First Backend Capstone Project Advance

RestTemplate, Resilience & Handling Failures in AI/API Systems

UUIDs and Representing Inheritance

Mocking and Writing Unit Tests (including AI services)

Advanced Testing Strategies for Backend Systems

JWT Generation and Validation

Authentication - Implementing OAuth 2

EBS, RDS

VPC, Security Groups, Route 53

Implementing Search: Paging, Sorting & Intro to Semantic Search

Creating Payment Microservice

Implementing Stripe PG and Reconciliation

Optimizing APIs using Redis and Kafka for Async & AI Workloads

Spring Cloud for Distributed Microservices & AI Services

Spring Cloud: API Gateway, Load Balancing, Logging & Monitoring

Docker for Backend & AI Service Deployment

Elective Build a Production-Ready AI-Enabled System

Module Introduction - Intro to LLMs for Backend Engineers & Prompt

Engineering for Backend APIs

Building BookMyShow Chat Agent - 1

Building BookMyShow Chat Agent - 2

Building BookMyShow Chat Agent - 3

Buildng Splitwise AI Receipt Processor - 1

Buildng Splitwise AI Receipt Processor - 2

Building Splitwise AI Receipt Processor - 3

Backend Capstone Project: Contest

Fullstack ●

Web Platform Foundations & UI Systems

How the Web Works: Browsers, HTML, Developer Tools, and AI Inspection

Semantic HTML, Forms, and Structured User Input

CSS Foundations: Cascade, Selectors, Box Model, and Units

Layout Systems I: Display, Positioning, and Flexbox

Layout Systems II: Responsive Design, Media Queries, and Grid

CSS Variables, Transitions, and Visual Polish

UI Debugging: Specificity, Inheritance, Stacking, DevTools, and AI Review

From Design to Code: AI-Assisted UI Build, Review, and Refinement

HTML & CSS Certification Contest

JavaScript Runtime Systems, Browser APIs & Product Engineering Labs

JavaScript Foundations: Values, Types, Variables, Control Flow, and Basic Programming

Execution Model: Scope, Hoisting, and How JavaScript Runs Closures, Lexical Scope, and Persistent State

Runtime Reasoning Lab: Predicting Output, Tracing Execution, and Debugging

Functional JavaScript: Functions, Callbacks, and Higher Order Thinking

The Browser as a Runtime: DOM Structure and Browser APIs

Event Systems: User Interaction, Handlers, and UI Behavior

Product Lab I: Realtime Weather App, JSON, APIs, Async Flow, and Rendering

Product Lab II: Kanban Board, Product Breakdown, UI Structure, and State Model

Product Lab III: Kanban Board, DOM Manipulation and Dynamic Rendering

Product Lab IV: Kanban Board, Business Logic, Persistence, and Local Storage

Product Lab V: Kanban Board, Edge Cases, UX Refinement, and Extensibility

Async Systems I: Callbacks, Web APIs, and Non Blocking Execution

Async Systems II: Promises, Microtasks, and Chained Control Flow

Async Systems III: Async Await, Error Handling, and Resilient Flows

Async Systems IV: Promise Combinators, Coordination, and Real World Async Patterns

Async Debugging Lab: Race Conditions, Failure States, and AI Assisted Investigation

Browser Event Flow: Bubbling, Capturing, and Propagation

Interaction Patterns Lab: Event Delegation, Scalable UI Behavior, and Machine Coding

Performance Patterns Lab: Debouncing, Throttling, and Efficient Frontend Flows

Object Model I: this, Method Context, and Function Invocation

JS Certification Contest

React and Frontend Product Architecture Fundamentals

React Foundations: Components, JSX and Thinking in UI Trees

React Tooling and TypeScript Setup: Build Tools, JSX, TS Basics and Project Structure

Props, Lists, Forms and Typed Component Composition

State, Events and Interactive UI in React

Lifting State Up, useEffect and Data Flow in Component Trees

Product Lab 1: IMDB App: App Setup, Routing and Frontend Structure

Product Lab 2: IMDB App: Movies Page, Pagination and Data Rendering

Product Lab 3: IMDB App : API Integration, Watchlist and Local Storage

Product Lab 4: IMDB App : Search, Sorting, Filtering and Shared State

Context API, State Sharing and Component Communication

Redux Foundations: Store Design, Actions and Predictable State

Redux Integration Lab: Product Scale State Management in React

React Certification Contest

Backend Product Systems, APIs and Production Engineering

Backend Foundations: Node.js, APIs, Express and Service Thinking

Data Systems: MongoDB, Mongoose, Schema Design and SQL vs NoSQL Thinking

Middleware, Validation and MVC for Scalable Backend Structure

Product Lab 1: Backend Architecture, Domain Modeling and Project Setup

Product Lab 2: Authentication, Authorization and Role Based Access

Product Lab 3: Protected Routes, Tokens and Session Aware Flows

Product Lab 4: Movies, Theatres and Show APIs

Product Lab 5: Partner Workflows, Admin Approval and Moderated Actions

Product Lab 6: Booking Flow, Seat Management and Transaction Safety

Product Lab 7: Payments, Ticket Generation and Order Lifecycle

Product Lab 8: Email Workflows, Password Reset and User Communication

Realtime Backend Systems: WebSockets, Live Updates and Event Driven Flows

MERN Certification Contest

Elective Backend Product Systems - Deep Dive

Backend Quality Lab: API Testing, Debugging and Error Handling

Backend Security Lab: Validation, Auth Hardening, Abuse Prevention and Safe Defaults

Production Readiness: Environment Management, Deployment, Logging and Docker Basics

Scalability Lab: Caching, Queues, Background Jobs and Data Flow Trade Offs

AI Feature Integration Lab: LLM APIs, Structured Responses and Backend Orchestration

Architecture and Interview Lab: Backend Design, Trade Offs and AI Assisted Review

Elective Foundations of AI-Assisted Full-Stack Development

Full-Stack Development in the Age of AI

AI Coding Tools for Full-Stack Builders

Prompting for UI, Logic, APIs, and Debugging

Reviewing and Validating AI-Generated Full-Stack Code

From Screen to Server: AI-Assisted Development Workflows

Building Faster Without Losing Fundamentals

Elective JavaScript Object System & Engineering Patterns

Object Model II: Classes, Constructors, and Object Oriented Modeling

Object Model III: Prototypes, Inheritance, and Reusable Behavior

JavaScript Engineering Lab: Call, Apply, Bind, and Polyfill Thinking

Polyfills Lab: Higher Order Methods and Implementation Reasoning

Data Handling Lab: Deep Copy, Shallow Copy, and Reference Behavior

Elective React, TypeScript and Frontend Product Architecture Deep Dive

TypeScript Foundations for JavaScript Developers

Advanced Hooks: useRef, useMemo, useCallback and useReducer

Custom Hooks and Reusable Logic Patterns

React Performance Lab: Rendering, Memoization and UI Efficiency

Machine Coding Lab: Component Design, State Design and Real UI Problem Solving

Architecture Studio: React Patterns, State Trade Offs and AI Assisted Review

Next.js Foundations: App Router, Layouts and Modern React Delivery

Next.js Data Fetching: Server and Client Components, Rendering Strategy and SEO

Next.js Product Lab: Forms, Route Handlers, Auth Flow and Deployment

Advanced React Systems: Fiber, Concurrent Rendering, Suspense, Lazy Loading and Error Boundaries

React Interview and Frontend Reasoning Lab

Forward Deployed Engineer (FDE)

FDE Foundations: Python, Workflow & Delivery

The FDE Role: Customer-Embedded Engineering

Problem Discovery & Solution Thinking for FDEs

Python Toolchain & Environment Setup (pyenv, uv, ruff, mypy)

Iterators, Generators & Comprehensions

Closures, Decorators & Context Managers

OOP, Concurrency & Async Python

Pydantic v2, Structured Logging & Profiling

Git Internals, GitHub Workflows & AI Dev Tools (Cursor, Claude Code, MCP)

Testing: pytest, Property-Based Testing & CI

Software Engineering Best Practices (SOLID, Hexagonal Architecture)

Linux, Shell & Scripting for FDEs

Assessment: Python Engineering Challenge

Backend Engineering, Observability & Advanced Data Systems

Introduction to Backend Engineering — HTTP, REST & API Design

FastAPI: Intro to Deep Dive

Auth: OAuth2, JWT, OIDC & RBAC

Building an Order Management Module & its API

Building Payments & Notifications (Stripe, Celery)

Building the Demo Frontend

Real-Time, Queues & Caching (WebSockets, SSE, Redis)

Observability & Distributed Tracing (OpenTelemetry, Grafana)

Postgres Advanced: JSONB, Full-Text Search & pgvector

Postgres Internals: EXPLAIN ANALYZE, Indexing & MVCC

NoSQL, Document DBs & Vector Databases

Dimensional Modeling, dbt & Modern Data Stack

Workflow Orchestration with Airflow & Data Quality

Event-Driven Architecture & Kafka Fundamentals

Assessment: Data Pipeline Build & Debug

Full-Stack FDE: TypeScript, React & AI-First Frontends

TypeScript Fundamentals for Python Engineers

React Essentials for AI Apps

Next.js for Full-Stack AI Apps

Vercel AI SDK & Streaming UIs

AI Product UX & Customer-Facing Polish

Cloud, DevOps, Kubernetes & Infrastructure

AWS Core: IAM, VPC & Networking

Compute, Storage & Managed Services (EC2, S3, RDS, SQS)

AWS Security & Secrets Management (KMS, Secrets Manager, CloudTrail)

Docker: Production Containers

Kubernetes Architecture & Core Concepts

EKS, Helm & Production Kubernetes

Kubernetes Networking, Security & GPU Workloads

CI/CD Pipelines & GitOps (GitHub Actions, ArgoCD)

Terraform & Infrastructure as Code

Serverless & Event-Driven Cloud (Lambda, API Gateway, Step Functions)

SRE & Reliability Engineering for FDEs

Assessment: Deploy, Break, Fix on Cloud

Enterprise Communication, Consulting & LLM Engineering

Customer Discovery & Requirements Gathering

Technical Scoping & Solution Documentation (PRD-lite, options matrix, risk register)

Stakeholder Communication & Written Artifacts

Demo Storytelling & Handling Pushback

LLM Fundamentals for Production

Prompt Engineering for Production Systems

RAG Architecture: Ingestion & Retrieval

RAG over Customer Documents & Hybrid Search

RAG Evaluation & Iteration (RAGAS)

LLM Safety, Guardrails & Security

Cost, Latency & Model Routing

Assessment: Discovery Call + RAG PoC Build

Agentic Systems & Enterprise Integrations

Tool Calling, Function Calling & Structured Output

Agentic Patterns & Production Concerns

Agent Debugging, Failure Analysis & Evaluation

MCP & Enterprise Tooling

Multi-Agent Patterns & Human-in-the-Loop

Assessment: Agentic System Build & Red-Team

Enterprise Systems Landscape & Integration Topology

API Integration Patterns & Connector Design

Integration Failure Modes & Debugging

Messy Customer Data & Entity Resolution

Build: CRM-to-AI Support Copilot

Assessment: Integration War Room

Application Engineering, Security & Reliability

Product Thinking & Architecture for FDE Apps

Frontend Essentials for FDEs: React + TypeScript

Multi-Tenant Design & Background Workflows (Postgres RLS, Celery)

Document Processing & AI Extraction Workflows

Build: Enterprise Document Review App

Developer Experience & API Documentation

Enterprise Security for FDE Systems (PII, SOC2, GDPR)

Secrets, Encryption & Deployment Topologies

Load Testing, Capacity Planning & Performance (k6)

Incident Response & Customer-Facing RCA

Enterprise Readiness Checklist & Go-Live

Assessment: Secure Deploy + Incident Simulation

System Design, Capstone & Hiring Readiness

Distributed Systems Fundamentals (CAP, PACELC, Raft, Saga)

System Design: RAG System at Scale

System Design: Agentic Platform at Scale

System Design: Multi-Tenant SaaS Platform

System Design: Enterprise Integration Hub

System Design Interview Framework & Practice

Capstone Sprint 1: Discovery & Solution Design

Capstone Sprint 2: Core Build & Integration Layer

Capstone Sprint 3: Harden, Secure & Observe

Capstone Sprint 4: Red-Team, Eval & Fix

Capstone Demo Day: Client Panel Presentation

Interview Readiness & Career Launch

FDE Mock Engagement: Full Simulation

Graduation Review: Portfolio Defense & Next Steps

Elective Containerization and Orchestration

Docker introduction

Docker image and Docker containers

Building your own Docker image

Image and Containers Deep Dive

Docker volume

Docker networking deep dive

Hardening the Docker env. (DevSecOps)

Demo of implementing industry standard implementation (e.g. in image hardening and other security standards)

Introduction to Orchestration

Kubernetes architecture

Kubernetes Pod deep dive

Kubernetes Replicaset and Deployment deep dive, side car

Kubernetes networking

Scaling options in kubernetes - HPA, CA and VPA - compare with

Karpentor

Introduction to scaling in kubernetes, Network policies

Kubernetes security - RBAC (DevSecOps)

Kubernetes volumes

Elective AWS : multi-region cloud deep dive

AWS VPC & Networking Basics

Advanced AWS Networking

AWS Observability & CloudWatch

CloudWatch & CloudTrail Advanced (DevSecOps)

Docker App on EC2 Project

AWS Lambda & API Gateway

End-to-End Project day

AWS ECS & ECR Intro

ECS Fundamental

EKS Basics

EKS Advanced

EKS Security & Operations

Microservices on AWS Project

Microservices Project – Phase 2

AWS Security: Identity & Network (DevSecOps)

AWS Security: Data & Detection (DevSecOps)

AWS Security Hands-on (DevSecOps)

Elective Storage, Networking & Reliability for FDEs- storage architecture, DR, sharding, CDN, HA, fault tolerance

Architect the Storage for an Application

Data Lifecycle Management and Storage Optimization

Backup and Disaster Recovery Strategies

Designing the Networking of an Application

Load Balancing and Auto Scaling Groups

Designing for scalability of application and data

Database Sharding Strategies and Hot-Spot Mitigation

Edge Caching, CDNs, and Latency Reduction

Microservices Performance — Latency, Throughput, Concurrency

Designing for High Availability — Multi-Region Active-Active

Fault Tolerance Engineering — Circuit Breakers, Retries, Backoff

Graceful Degradation — Rate Limiting, Throttling, Shedding

Data Consistency Trade-offs — CAP, PACELC

System Design ●

Distributed System Design & AI-Integrated Architectures

System Design Foundations & Modern Infrastructure

Load Balancing, Consistent Hashing & Traffic Routing at Scale

Caching Systems: CDN, Backend Caches, Cache Invalidation

Case Study: Caching at Scale (Scaler Code Judge and Contest Leaderboards + AI Workloads)

Case Study: Caching Facebook News Feed

CAP / PACELC theorem + Master Slave Replication

SQL vs NoSQL + Sharding

Database Orchestration & Shard Creation

Case study: Google Typeahead (How to approach System Design Problems)

Case Study: Google Typeahead (Design and Optimizations)

NoSQL Internals - LSM Tree and Multi Master

Case Study: Messaging Apps (FB Messenger, Whatsapp, Slack)

Messaging Queues - Apache Kafka & Zookeeper

Case Study: Elasticsearch (Full Text Search)

Case Study: S3, HDFS (Large File Storage)

Case Study: Uber (Nearest Neighbor Search)

Case Study: Rate Limiter + Unique ID Generator (Infra)

Case Study: OTT Platform (OTT)

Microservices - 1

Microservices - 2

Case Study: Ecommerce Platform (Microservices)

Interview Preparation ●

Mastering LastMile - Interview Simulation

DSA Interview Simulation: Problem Solving & Thinking Under Pressure

LLD Interview Simulation: Structuring Scalable Solutions

HLD Interview Simulation: Designing Systems & Communicating Trad

Data Engineering ●

Data Engineering

Introduction to Data Engineering

Writing Efficient Queries: Part 1

Writing Efficient Queries: Part 2

Intro to Hadoop

HDFS and MAP-REDUCE

Introduction to Hive

Advanced features of Hive

Basics of Spark

Spark Programming

Structured API - PySpark

PySpark Operations

Introduction to Streaming

Apache Spark Streaming

Advanced Spark Streaming

Additional Streaming Services: Bringing it all together

Modern Data Architectures

AWS Services for Data Engineering

Data Modelling

Orchestrating data pipelines

Orchestration Continued

Distributed Coordination

Load Balancing

Data Pillars

Capstone

Advanced DSA ●

DSA Edge: Advanced Patterns & Interview Techniques

DSA Edge : Townhall

DSA Edge : Townhall (Post-SQL - change name after scheduling)

Revision of DSA Topics

Maths: Inverse Mod & Problems

Backtracking: Famous Problems

Tries 1: Trie of Character

Tries 2: Trie of Bits + Problems on Trees

Strings Pattern Matching

DP 1: DP on Strings

DP 2: Famous Problems

Graphs 1: Bipartite Graph

Graphs 2: Belleman Ford & Floyd Warshall Algorithm

Advanced Interview Problems

DSA Edge Mandatory Contest: Full Syllabus

DSA Edge Full Syllabus: Contest Discussion

Product Management for Engineers

Townhall: Product Management

Introduction to Product Management 1

Understanding the Product Lifecycle 1

Understanding the Product Lifecycle 2

Critical Thinking in Product Development 1

Critical Thinking in Product Development 2

Market Structure Analysis and Its Technical Aspects 1

Market Structure Analysis and Its Technical Aspects 2

MVPs, Prototyping, and Growth Hacking 1

MVPs, Prototyping, and Growth Hacking 2

Concept Development, Validation, and the Role of Developers 1

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Refresher: Modern Text & Image AI Model Architectures

Getting started with Text Generation LLM APIs (OpenAI API)

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Prompt Engineering: Introduction

Retrieval-Augmented Generation (RAG) Introduction

Embeddings Deep Dive

Multimodal & Tabular RAG

Prompt Engineering: Security

Agents: Foundations & Planning

Advanced Agent Concepts

Agent: Frameworks and Protocols

Fine Tuning Existing Models

Audio, Image, Video, Expressions: AI Modalities

Scaling AI Applications

Model Quantization Techniques

Advanced Fine Tuning Techniques

Dataset Engineering & Inference Optimization

Intermediate

Programming Fundamentals ●

Intermediate DSA : AI Assisted Problem Solving

Time Complexity

Introduction to Arrays & Basics of AI

Lab Session on TC, SC, Output & Debugging

Arrays: Prefix Sum & Carry Forward with AI-Assisted Problem Solving

Lab Session on Prefix Sum & Carry Forward

Arrays: Arrays techniques & smart prompting

Lab Session on Memory Management & Sorting Basics

Bit Manipulations Basics

Lab Session on 2D Matrices

Strings & Immutability visualisations through AI

Lab on Matrix & Strings using AI

Contest

Agentic AI ●

AI & Agents: From Talking to AI to Building One

GenAI World & LLM Landscape

Prompt Engineering Basics

Advanced Prompting & AI Safety Basics

LLM Built-in Power Tools

RAG

n8n Automation Workflow

What Is an AI Agent?

Building Agents with n8n

OpenClaw: Personal 24/7 AI Agent

Multi-Agent Systems

Debugging AI

Advanced DSA ●

Advanced DSA: Foundations:Core Techniques & Optimization

Time Complexity

Arrays Techniques

Arrays 1: One Dimensional

Arrays 2: Two Dimensional

Lab Session on Arrays

Bit Manipulation

Lab Session on Bit Manipulation

Recursion

Lab Session on Recursion

Maths: Modular Arithmetic & GCD

Hashing

Lab Session on Hashing

Sorting 1: Count Sort & Merge Sort

Sorting 2: Quick Sort & Comparator Problems

Sample Contest: Dual Camera Proctoring

Contest 1: Arrays, Bit Manipulation, Recursion, Math, Hashing & Sorting

Advanced DSA: Linear & Non-Linear Structures

Searching 1: Binary Search on Array

Searching 2: Binary Search on Answer

Lab Session on Searching

Classes, Objects & Linked List Introduction

Linked List: Basic Problems

Stacks

Lab Session on Stacks

Queues: Implementation & Problems

Trees 1: Structure & Traversal

Trees 2: BST

Lab Session on Binary Trees

Revision of DSA 1 & 2

Break

Contest 2 : Searching, Linked List, Stacks, Queues , Trees & contest discussion

Advanced DSA: Backtracking & Advanced Trees

Maths: Combinatorics Basics & Prime Numbers

Lab Session on Prime Numbers & 2 Pointers

Lab Session on Maths & 2 Pointers

Backtracking

Lab Session on Backtracking

Linked List: Sorting and Problems

Linked List: Doubly Linked List & Detecting Loop

Trees 4: Morris Inorder Traversal + LCA

Lab Session on Binary Trees 2

Hashing 3: Internal Implementation & Problems

Contest 3: Math, Two Pointers, Backtracking, Linked List , Trees & contest discussion

Advanced DSA: DP, Heaps & Graphs — Crack It with AI Edge

Heaps: Introduction

Heap Sort & Greedy

Lab Session on Heaps & Greedy

Lab Session on Interview Problems 1

DP 1: One Dimensional

DP 2: Two Dimensional

DP 3: Knapsack

Lab Session on Applications of Knapsack

Graphs 1: Introduction, DFS & Cycle Detection

Graphs 2: BFS & MST

Graphs 3: Dijkstra Algo & Topological Sort

Lab Session on Interview Problems 2

Revision of DSA 3 & 4

Contest 4: Heaps, Greedy, DP & Graphs

Mandatory Skill Evaluation Test: DSA

Skill Evaluation Test Discussion + How to ace DSA Interview

DSA : Real World Project 1 (Social Network Analyzer)

DSA : Real world Project 2 (Invoice Merger)

Databases & SQL ●

Databases & SQL

Introduction to DBMS & Keys

CRUD Operations - 1

CRUD Operations - 2

Joins

Aggregate Queries

Joins and Aggregate Queries Working Session

Subqueries & Views

Indexing

Transactions

Window Functions

Schema Design - 1

Schema Design - 2

Databases & SQL Contest: SQL and Schema Design

Backend ●

Foundations of Object-Oriented Design & Scalable Systems

Intro to LLD for Scalable & AI-Ready Systems

OOP-1: Intro to OOP Lab

OOP-2: Access Modifiers and Constructors

OOP-3: Inheritance and Polymorphism

OOP-4: Interfaces, Abstract Classes

Concurrency-1: Introduction to Processes and Threads

Concurrency-2: Executors and Callables

Concurrency-3: Introduction to Synchronization

Concurrency-4: Synchronization with Semaphores

Java Advanced Concepts - 1 [Generics]

Java Advanced Concepts - 2 [Generics & Collections Lab]

Java Advanced Concepts - 3 [Streams and Lambdas]

Java Advanced Concepts - 4 [Exception Handling and Miscellaneous Topics]

Contest - 1: Java, OOP, and Concurrency

Design Principles & Patterns for Extensible Systems

SOLID Principles for Extensible Systems with AI integrations

Design Patterns: Introduction and Singleton

Design Patterns: Builder

Design Patterns: Prototype and Registry

Design Patterns: Leveraging Factory Pattern for Service Abstraction (APIs & AI Services)

Design Patterns: Adapter and Facade

Design Patterns: Decorator and Flyweight

Design Patterns: Behavioural Design Patterns

UML Diagrams

Contest - 2: Design Principles & Design Patterns

Applied Design & Machine Coding

How to Approach Design Problems

Design TicTacToe

Code TicTacToe 1

Code TicTacToe 2

Design Parking Lot

Code Parking Lot 1

Design BookMyShow

Code BookMyShow 1

Code BookMyShow 2

Code BookMyShow 3

Design Splitwise

Code Splitwise 1

Code Splitwise 2

Machine Coding

AI-First Backend Capstone Project Primer

Intro to Backend Systems & AI-Augmented Development

Version Control and Git

Learning Git Commands & AI Dev Tools

Intro to Spring Framework and Building APIs

MVC, Requests in Spring and Starting First Microservice (with AI Service Integration)

Calling 3rd Party APIs & LLM/AI Services

Intro to Spring Data JPA

JPA Queries and Repositories

Unit Testing and Good Practices (including AI-assisted testing)

Authentication vs Authorization, OAuth2, JWT

Implementing UserAuthService

Deploying Backend & AI-Enabled Applications to AWS

AI-First Backend Capstone Project Advance

RestTemplate, Resilience & Handling Failures in AI/API Systems

UUIDs and Representing Inheritance

Mocking and Writing Unit Tests (including AI services)

Advanced Testing Strategies for Backend Systems

JWT Generation and Validation

Authentication - Implementing OAuth 2

EBS, RDS

VPC, Security Groups, Route 53

Implementing Search: Paging, Sorting & Intro to Semantic Search

Creating Payment Microservice

Implementing Stripe PG and Reconciliation

Optimizing APIs using Redis and Kafka for Async & AI Workloads

Spring Cloud for Distributed Microservices & AI Services

Spring Cloud: API Gateway, Load Balancing, Logging & Monitoring

Docker for Backend & AI Service Deployment

Elective Build a Production-Ready AI-Enabled System

Module Introduction - Intro to LLMs for Backend Engineers & Prompt

Engineering for Backend APIs

Building BookMyShow Chat Agent - 1

Building BookMyShow Chat Agent - 2

Building BookMyShow Chat Agent - 3

Buildng Splitwise AI Receipt Processor - 1

Buildng Splitwise AI Receipt Processor - 2

Buildng Splitwise AI Receipt Processor - 3

Backend Capstone Project: Contest

Fullstack ●

Web Platform Foundations & UI Systems

How the Web Works: Browsers, HTML, Developer Tools, and AI Inspection

Semantic HTML, Forms, and Structured User Input

CSS Foundations: Cascade, Selectors, Box Model, and Units

Layout Systems I: Display, Positioning, and Flexbox

Layout Systems II: Responsive Design, Media Queries, and Grid

CSS Variables, Transitions, and Visual Polish

UI Debugging: Specificity, Inheritance, Stacking, DevTools, and AI Review

From Design to Code: AI-Assisted UI Build, Review, and Refinement

HTML & CSS Certification Contest

JavaScript Runtime Systems, Browser APIs & Product Engineering Labs

Full Stack LLD: JS-1: JS Refresher and Code Execution

Full Stack LLD: JS-2: OOPS-1 : This, Bind, Call, Apply, Inheritance

Full Stack LLD: JS-3: Polyfills of call,bind ,apply & deep copy-shallow copy

"Full Stack LLD: JS-4: Array , HOFs and it's Polyfills"

Full Stack LLD: JS-5: ES6 Classes and Objects

Full Stack LLD: JS-6: Closure and It's application

Full Stack LLD: JS-7: OOPS-2 : Object creation, Freezing objects

Full Stack LLD: JS-8: Event loop and Callback

Full Stack LLD: JS-9: Promises and MicroTask Queue

Full Stack LLD: JS-10: Promise chaining and Imp polyfills

Full Stack LLD: JS-11: Async await & Error handling

Full Stack LLD: JS-12: ES6 DataTypes and Modules

Full Stack LLD: FE Machine coding-1:Intro to the DOM

Full Stack LLD: FE Machine coding-2: Events & Event Handling, Bubbling & capturing

Full Stack LLD & Projects Certification Contest - JavaScript

React and Frontend Product Architecture Fundamentals

React Foundations: Components, JSX and Thinking in UI Trees

React Tooling and TypeScript Setup: Build Tools, JSX, TS Basics and Project Structure

Props, Lists, Forms and Typed Component Composition

State, Events and Interactive UI in React

Lifting State Up, useEffect and Data Flow in Component Trees

Product Lab 1: IMDB App: App Setup, Routing and Frontend Structure

Product Lab 2: IMDB App: Movies Page, Pagination and Data Rendering

Product Lab 3: IMDB App : API Integration, Watchlist and Local Storage

Product Lab 4: IMDB App : Search, Sorting, Filtering and Shared State

Context API, State Sharing and Component Communication

Redux Foundations: Store Design, Actions and Predictable State

Redux Integration Lab: Product Scale State Management in React

React Certification Contest

Backend Product Systems, APIs and Production Engineering

Backend Foundations: Node.js, APIs, Express and Service Thinking

Data Systems: MongoDB, Mongoose, Schema Design and SQL vs NoSQL Thinking

Middleware, Validation and MVC for Scalable Backend Structure

Product Lab 1: Backend Architecture, Domain Modeling and Project Setup

Product Lab 2: Authentication, Authorization and Role Based Access

Product Lab 3: Protected Routes, Tokens and Session Aware Flows

Product Lab 4: Movies, Theatres and Show APIs

Product Lab 5: Partner Workflows, Admin Approval and Moderated Actions

Product Lab 6: Booking Flow, Seat Management and Transaction Safety

Product Lab 7: Payments, Ticket Generation and Order Lifecycle

Product Lab 8: Email Workflows, Password Reset and User Communication

Realtime Backend Systems: WebSockets, Live Updates and Event Driven Flows

Backend Product Systems - Deep Dive

Backend Quality Lab: API Testing, Debugging and Error Handling

Backend Security Lab: Validation, Auth Hardening, Abuse Prevention and Safe Defaults

Production Readiness: Environment Management, Deployment, Logging and Docker Basics

Scalability Lab: Caching, Queues, Background Jobs and Data Flow Trade Offs

AI Feature Integration Lab: LLM APIs, Structured Responses and Backend Orchestration

Architecture and Interview Lab: Backend Design, Trade Offs and AI Assisted Review

MERN Certification Contest

Elective Foundations of AI-Assisted Full-Stack Development

Full-Stack Development in the Age of AI

AI Coding Tools for Full-Stack Builders

Prompting for UI, Logic, APIs, and Debugging

Reviewing and Validating AI-Generated Full-Stack Code

From Screen to Server: AI-Assisted Development Workflows

Building Faster Without Losing Fundamentals

Elective Frontend Systems & Machine Coding Mastery

Full Stack LLD: FE Machine coding: Case Studies

Full Stack LLD: FE Machine coding: Browser Perf & Memory leaks

Full Stack LLD: FE Machine coding: HTTP and network Optimization

Full Stack LLD: FE Machine coding:- Typeahead[Intersection Obs, Debouncing, Throttling]

Full Stack LLD: FE Machine coding: - Kanban board-1

Full Stack LLD: FE Machine coding: Kanban board-2

Full Stack LLD & Projects: Contest - FE Machine Coding

Elective React, TypeScript and Frontend Product Architecture Deep Dive

TypeScript Foundations for JavaScript Developers

Advanced Hooks: useRef, useMemo, useCallback and useReducer

Custom Hooks and Reusable Logic Patterns

React Performance Lab: Rendering, Memoization and UI Efficiency

Machine Coding Lab: Component Design, State Design and Real UI Problem Solving

Architecture Studio: React Patterns, State Trade Offs and AI Assisted Review

Next.js Foundations: App Router, Layouts and Modern React Delivery

Next.js Data Fetching: Server and Client Components, Rendering Strategy and SEO

Next.js Product Lab: Forms, Route Handlers, Auth Flow and Deployment

Advanced React Systems: Fiber, Concurrent Rendering, Suspense, Lazy Loading and Error Boundaries

React Interview and Frontend Reasoning Lab

Forward Deployed Engineer (FDE) ●

FDE Foundations: Python, Workflow & Delivery

The FDE Role: Customer-Embedded Engineering

Problem Discovery & Solution Thinking for FDEs

Python Toolchain & Environment Setup (pyenv, uv, ruff, mypy)

Iterators, Generators & Comprehensions

Closures, Decorators & Context Managers

OOP, Concurrency & Async Python

Pydantic v2, Structured Logging & Profiling

Git Internals, GitHub Workflows & AI Dev Tools (Cursor, Claude Code, MCP)

Testing: pytest, Property-Based Testing & CI

Software Engineering Best Practices (SOLID, Hexagonal Architecture)

Linux, Shell & Scripting for FDEs

Assessment: Python Engineering Challenge

Backend Engineering, Observability & Advanced Data Systems

Introduction to Backend Engineering — HTTP, REST & API Design

FastAPI: Intro to Deep Dive

Auth: OAuth2, JWT, OIDC & RBAC

Building an Order Management Module & its API

Building Payments & Notifications (Stripe, Celery)

Building the Demo Frontend

Real-Time, Queues & Caching (WebSockets, SSE, Redis)

Observability & Distributed Tracing (OpenTelemetry, Grafana)

Postgres Advanced: JSONB, Full-Text Search & pgvector

Postgres Internals: EXPLAIN ANALYZE, Indexing & MVCC

NoSQL, Document DBs & Vector Databases

Dimensional Modeling, dbt & Modern Data Stack

Workflow Orchestration with Airflow & Data Quality

Event-Driven Architecture & Kafka Fundamentals

Assessment: Data Pipeline Build & Debug

Full-Stack FDE: TypeScript, React & AI-First Frontends

TypeScript Fundamentals for Python Engineers

React Essentials for AI Apps

Next.js for Full-Stack AI Apps

Vercel AI SDK & Streaming UIs

AI Product UX & Customer-Facing Polish

Cloud, DevOps, Kubernetes & Infrastructure

AWS Core: IAM, VPC & Networking

Compute, Storage & Managed Services (EC2, S3, RDS, SQS)

AWS Security & Secrets Management (KMS, Secrets Manager, CloudTrail)

Docker: Production Containers

Kubernetes Architecture & Core Concepts

EKS, Helm & Production Kubernetes

Kubernetes Networking, Security & GPU Workloads

CI/CD Pipelines & GitOps (GitHub Actions, ArgoCD)

Terraform & Infrastructure as Code

Serverless & Event-Driven Cloud (Lambda, API Gateway, Step Functions)

SRE & Reliability Engineering for FDEs

Assessment: Deploy, Break, Fix on Cloud

Enterprise Communication, Consulting & LLM Engineering

Customer Discovery & Requirements Gathering

Technical Scoping & Solution Documentation (PRD-lite, options matrix, risk register)

Stakeholder Communication & Written Artifacts

Demo Storytelling & Handling Pushback

LLM Fundamentals for Production

Prompt Engineering for Production Systems

RAG Architecture: Ingestion & Retrieval

RAG over Customer Documents & Hybrid Search

RAG Evaluation & Iteration (RAGAS)

LLM Safety, Guardrails & Security

Cost, Latency & Model Routing

Assessment: Discovery Call + RAG PoC Build

Agentic Systems & Enterprise Integrations

Tool Calling, Function Calling & Structured Output

Agentic Patterns & Production Concerns

Agent Debugging, Failure Analysis & Evaluation

MCP & Enterprise Tooling

Multi-Agent Patterns & Human-in-the-Loop

Assessment: Agentic System Build & Red-Team

Enterprise Systems Landscape & Integration Topology

API Integration Patterns & Connector Design

Integration Failure Modes & Debugging

Messy Customer Data & Entity Resolution

Build: CRM-to-AI Support Copilot

Assessment: Integration War Room

Application Engineering, Security & Reliability

Product Thinking & Architecture for FDE Apps

Frontend Essentials for FDEs: React + TypeScript

Multi-Tenant Design & Background Workflows (Postgres RLS, Celery)

Document Processing & AI Extraction Workflows

Build: Enterprise Document Review App

Developer Experience & API Documentation

Enterprise Security for FDE Systems (PII, SOC2, GDPR)

Secrets, Encryption & Deployment Topologies

Load Testing, Capacity Planning & Performance (k6)

Incident Response & Customer-Facing RCA

Enterprise Readiness Checklist & Go-Live

Assessment: Secure Deploy + Incident Simulation

System Design, Capstone & Hiring Readiness

Distributed Systems Fundamentals (CAP, PACELC, Raft, Saga)

System Design: RAG System at Scale

System Design: Agentic Platform at Scale

System Design: Multi-Tenant SaaS Platform

System Design: Enterprise Integration Hub

System Design Interview Framework & Practice

Capstone Sprint 1: Discovery & Solution Design

Capstone Sprint 2: Core Build & Integration Layer

Capstone Sprint 3: Harden, Secure & Observe

Capstone Sprint 4: Red-Team, Eval & Fix

Capstone Demo Day: Client Panel Presentation

Interview Readiness & Career Launch

FDE Mock Engagement: Full Simulation

Graduation Review: Portfolio Defense & Next Steps

Elective Containerization and Orchestration

Docker introduction

Docker image and Docker containers

Building your own Docker image

Image and Containers Deep Dive

Docker volume

Docker networking deep dive

Hardening the Docker env. (DevSecOps)

Demo of implementing industry standard implementation (e.g. in image hardening and other security standards)

Introduction to Orchestration

Kubernetes architecture

Kubernetes Pod deep dive

Kubernetes Replicaset and Deployment deep dive, side car

Kubernetes networking

Scaling options in kubernetes - HPA, CA and VPA - compare with Karpenter

Introduction to scaling in kubernetes, Network policies

Kubernetes security - RBAC (DevSecOps)

Kubernetes volumes

Elective AWS : multi-region cloud deep dive

AWS VPC & Networking Basics

Advanced AWS Networking

AWS Observability & CloudWatch

CloudWatch & CloudTrail Advanced (DevSecOps)

Docker App on EC2 Project

AWS Lambda & API Gateway

End-to-End Project day

AWS ECS & ECR Intro

ECS Fundamental

EKS Basics

EKS Advanced

EKS Security & Operations

Microservices on AWS Project

Microservices Project – Phase 2

AWS Security: Identity & Network (DevSecOps)

AWS Security: Data & Detection (DevSecOps)

AWS Security Hands-on (DevSecOps)

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Data Engineering

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Orchestration Continued

Distributed Coordination

Load Balancing

Data Pillars

Capstone

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DSA Edge : Townhall

DSA Edge : Townhall (Post-SQL - change name after scheduling)

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Townhall: Product Management

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Understanding the Product Lifecycle 2

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Scaling AI Applications

Model Quantization Techniques

Advanced Fine Tuning Techniques

Dataset Engineering & Inference Optimization

Advanced

Agentic AI ●

AI & Agents: From Talking to AI to Building One

- The runway to AI
- GenAI World & LLM Landscape
- Prompt Engineering Basics
- Advanced Prompting & AI Safety Basics
- LLM Built-in Power Tools
- RAG
- n8n Automation Workflow
- What Is an AI Agent?
- Building Agents with n8n
- OpenClaw: Personal 24/7 AI Agent
- Multi-Agent Systems
- Debugging AI with Vibecoding Tools

Advanced DSA ●

Advanced DSA: Foundations:Core Techniques & Optimization

- Time Complexity
- Arrays Techniques
- Arrays 1: One Dimensional
- Arrays 2: Two Dimensional
- Lab Session on Arrays
- Bit Manipulation
- Lab Session on Bit Manipulation
- Recursion
- Lab Session on Recursion
- Maths: Modular Arithmetic & GCD
- Hashing
- Lab Session on Hashing
- Sorting 1: Count Sort & Merge Sort
- Sorting 2: Quick Sort & Comparator Problems
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- Contest 1: Arrays, Bit Manipulation, Recursion, Math, Hashing & Sorting

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Linked List: Basic Problems

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Lab Session on Stacks

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Lab Session on Prime Numbers & 2 Pointers

Lab Session on Maths & 2 Pointers

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Linked List: Doubly Linked List & Detecting Loop

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Contest 3: Math, Two Pointers, Backtracking, Linked List , Trees & contest discussion

Advanced DSA: DP, Heaps & Graphs — Crack It with AI Edge

Heaps: Introduction

Heap Sort & Greedy

Lab Session on Heaps & Greedy

Lab Session on Interview Problems 1

DP 1: One Dimensional

DP 2: Two Dimensional

DP 3: Knapsack

Lab Session on Applications of Knapsack

Graphs 1: Introduction, DFS & Cycle Detection

Graphs 2: BFS & MST

Graphs 3: Dijkstra Algo & Topological Sort

Lab Session on Interview Problems 2

Revision of DSA 3 & 4

Contest 4: Heaps, Greedy, DP & Graphs

Mandatory Skill Evaluation Test: DSA

Skill Evaluation Test Discussion + How to ace DSA Interview

DSA : Real World Project 1 (Social Network Analyzer)

DSA : Real world Project 2 (Invoice Merger)

Databases & SQL ●

Databases & SQL

Introduction to DBMS & Keys

CRUD Operations - 1

CRUD Operations - 2

Joins

Aggregate Queries

Joins and Aggregate Queries Working Session

Subqueries & Views

Indexing

Transactions

Window Functions

Schema Design - 1

Schema Design - 2

SQL and Schema Design

Backend ●

Foundations of Object-Oriented Design & Scalable Systems

Intro to LLD for Scalable & AI-Ready Systems

OOP-1: Intro to OOP Lab

OOP-2: Access Modifiers and Constructors

OOP-3: Inheritance and Polymorphism

OOP-4: Interfaces, Abstract Classes

Concurrency-1: Introduction to Processes and Threads

Concurrency-2: Executors and Callables

Concurrency-3: Introduction to Synchronization

Concurrency-4: Synchronization with Semaphores

Java Advanced Concepts - 1 [Generics]

Java Advanced Concepts - 2 [Generics & Collections Lab]

Java Advanced Concepts - 3 [Streams and Lambdas]

Java Advanced Concepts - 4 [Exception Handling and Miscellaneous Topics]

Contest - 1: Java, OOP, and Concurrency

Design Principles & Patterns for Extensible Systems

SOLID Principles for Extensible Systems with AI integrations

Design Patterns: Introduction and Singleton

Design Patterns: Builder

Design Patterns: Prototype and Registry

Design Patterns: Leveraging Factory Pattern for Service Abstraction (APIs & AI Services)

Design Patterns: Adapter and Facade

Design Patterns: Decorator and Flyweight

Design Patterns: Behavioural Design Patterns

UML Diagrams

Contest - 2: Design Principles & Design Patterns

Applied Design & Machine Coding

How to Approach Design Problems

Design TicTacToe

Code TicTacToe 1

Code TicTacToe 2

Design Parking Lot

Code Parking Lot 1

Design BookMyShow

Code BookMyShow 1

Code BookMyShow 2

Code BookMyShow 3

Design Splitwise

Code Splitwise 1

Code Splitwise 2

Low Level Design & Machine Coding

AI-First Backend Capstone Project Primer

Intro to Backend Systems & AI-Augmented Development

Version Control and Git

Learning Git Commands & AI Dev Tools

Intro to Spring Framework and Building APIs

MVC, Requests in Spring and Starting First Microservice (with AI Service Integration)

Calling 3rd Party APIs & LLM/AI Services

Intro to Spring Data JPA

JPA Queries and Repositories

Unit Testing and Good Practices (including AI-assisted testing)

Authentication vs Authorization, OAuth2, JWT

Implementing UserAuthService

Deploying Backend & AI-Enabled Applications to AWS

AI-First Backend Capstone Project Advance

RestTemplate, Resilience & Handling Failures in AI/API Systems

UUIDs and Representing Inheritance

Mocking and Writing Unit Tests (including AI services)

Advanced Testing Strategies for Backend Systems

JWT Generation and Validation

Authentication - Implementing OAuth 2

EBS, RDS

VPC, Security Groups, Route 53

Implementing Search: Paging, Sorting & Intro to Semantic Search

Creating Payment Microservice

Implementing Stripe PG and Reconciliation

Optimizing APIs using Redis and Kafka for Async & AI Workloads

Spring Cloud for Distributed Microservices & AI Services

Spring Cloud: API Gateway, Load Balancing, Logging & Monitoring

Docker for Backend & AI Service Deployment

Elective Build a Production-Ready AI-Enabled System

Module Introduction - Intro to LLMs for Backend Engineers & Prompt Engineering for Backend APIs	
Building BookMyShow Chat Agent - 1	
Building BookMyShow Chat Agent - 2	
Building BookMyShow Chat Agent - 3	
Buildng Splitwise AI Receipt Processor - 1	
Buildng Splitwise AI Receipt Processor - 2	
Buildng Splitwise AI Receipt Processor - 3	
Backend Capstone Project: Contest	

Fullstack ●

Web Platform Foundations & UI Systems

How the Web Works: Browsers, HTML, Developer Tools, and AI Inspection	
Semantic HTML, Forms, and Structured User Input	
CSS Foundations: Cascade, Selectors, Box Model, and Units	
Layout Systems I: Display, Positioning, and Flexbox	
Layout Systems II: Responsive Design, Media Queries, and Grid	
CSS Variables, Transitions, and Visual Polish	
UI Debugging: Specificity, Inheritance, Stacking, DevTools, and AI Review	
From Design to Code: AI-Assisted UI Build, Review, and Refinement	
HTML & CSS Certification Contest	

JavaScript Runtime Systems, Browser APIs & Product Engineering Labs

Full Stack LLD: JS-1: JS Refresher and Code Execution	
Full Stack LLD: JS-2: OOPS-1 : This, Bind, Call, Apply, Inheritance	
Full Stack LLD: JS-3: Polyfills of call,bind ,apply & deep copy-shallow copy	
"Full Stack LLD: JS-4: Array , HOFs and it's Polyfills"	
Full Stack LLD: JS-5: ES6 Classes and Objects	
Full Stack LLD: JS-6: Closure and It's application	
Full Stack LLD: JS-7: OOPS-2 : Object creation, Freezing objects	
Full Stack LLD: JS-8: Event loop and Callback	
Full Stack LLD: JS-9: Promises and MicroTask Queue	

Full Stack LLD: JS-10: Promise chaining and Imp polyfills

Full Stack LLD: JS-11: Async await & Error handling

Full Stack LLD: JS-12: ES6 DataTypes and Modules

Full Stack LLD: FE Machine coding-1: Intro to the DOM

Full Stack LLD: FE Machine coding-2: Events & Event Handling, Bubbling & capturing

Full Stack LLD & Projects Certification Contest - JavaScript

React and Frontend Product Architecture Fundamentals

React Foundations: Components, JSX and Thinking in UI Trees

React Tooling and TypeScript Setup: Build Tools, JSX, TS Basics and Project Structure

Props, Lists, Forms and Typed Component Composition

State, Events and Interactive UI in React

Lifting State Up, useEffect and Data Flow in Component Trees

Product Lab 1: IMDB App: App Setup, Routing and Frontend Structure

Product Lab 2: IMDB App: Movies Page, Pagination and Data Rendering

Product Lab 3: IMDB App : API Integration, Watchlist and Local Storage

Product Lab 4: IMDB App : Search, Sorting, Filtering and Shared State

Context API, State Sharing and Component Communication

Redux Foundations: Store Design, Actions and Predictable State

Redux Integration Lab: Product Scale State Management in React

React Certification Contest

Backend Product Systems, APIs and Production Engineering

Backend Foundations: Node.js, APIs, Express and Service Thinking

Data Systems: MongoDB, Mongoose, Schema Design and SQL vs NoSQL Thinking

Middleware, Validation and MVC for Scalable Backend Structure

Product Lab 1: Backend Architecture, Domain Modeling and Project Setup

Product Lab 2: Authentication, Authorization and Role Based Access

Product Lab 3: Protected Routes, Tokens and Session Aware Flows

Product Lab 4: Movies, Theatres and Show APIs

Product Lab 5: Partner Workflows, Admin Approval and Moderated Actions

Product Lab 6: Booking Flow, Seat Management and Transaction Safety

Product Lab 7: Payments, Ticket Generation and Order Lifecycle

Product Lab 8: Email Workflows, Password Reset and User Communication

Realtime Backend Systems: WebSockets, Live Updates and Event Driven Flows

Backend Product Systems - Deep Dive

Backend Quality Lab: API Testing, Debugging and Error Handling

Backend Security Lab: Validation, Auth Hardening, Abuse Prevention and Safe Defaults

Production Readiness: Environment Management, Deployment, Logging and Docker Basics

Scalability Lab: Caching, Queues, Background Jobs and Data Flow Trade Offs

AI Feature Integration Lab: LLM APIs, Structured Responses and Backend Orchestration

Architecture and Interview Lab: Backend Design, Trade Offs and AI Assisted Review

MERN Certification Contest

Elective Foundations of AI-Assisted Full-Stack Development

Full-Stack Development in the Age of AI

AI Coding Tools for Full-Stack Builders

Prompting for UI, Logic, APIs, and Debugging

Reviewing and Validating AI-Generated Full-Stack Code

From Screen to Server: AI-Assisted Development Workflows

Building Faster Without Losing Fundamentals

Elective Frontend Systems & Machine Coding Mastery

Full Stack LLD: FE Machine coding: Case Studies

Full Stack LLD: FE Machine coding: Browser Perf & Memory leaks

Full Stack LLD: FE Machine coding: HTTP and network Optimization

Full Stack LLD: FE Machine coding:- Typeahead[Intersection Obs, Debouncing, Throttling]

Full Stack LLD: FE Machine coding: - Kanban board-1

Full Stack LLD: FE Machine coding: Kanban board-2

Full Stack LLD & Projects: Contest - FE Machine Coding

Elective React, TypeScript and Frontend Product Architecture Deep Dive

TypeScript Foundations for JavaScript Developers

Advanced Hooks: useRef, useMemo, useCallback and useReducer

Custom Hooks and Reusable Logic Patterns

React Performance Lab: Rendering, Memoization and UI Efficiency

Machine Coding Lab: Component Design, State Design and Real UI Problem Solving

Architecture Studio: React Patterns, State Trade Offs and AI Assisted Review

Next.js Foundations: App Router, Layouts and Modern React Delivery

Next.js Data Fetching: Server and Client Components, Rendering Strategy and SEO

Next.js Product Lab: Forms, Route Handlers, Auth Flow and Deployment

Advanced React Systems: Fiber, Concurrent Rendering, Suspense, Lazy Loading and Error Boundaries

React Interview and Frontend Reasoning Lab

Forward Deployed Engineer (FDE) ●

FDE Foundations: Python, Workflow & Delivery

The FDE Role: Customer-Embedded Engineering

Problem Discovery & Solution Thinking for FDEs

Python Toolchain & Environment Setup (pyenv, uv, ruff, mypy)

Iterators, Generators & Comprehensions

Closures, Decorators & Context Managers

OOP, Concurrency & Async Python

Pydantic v2, Structured Logging & Profiling

Git Internals, GitHub Workflows & AI Dev Tools (Cursor, Claude Code, MCP)

Testing: pytest, Property-Based Testing & CI

Software Engineering Best Practices (SOLID, Hexagonal Architecture)

Linux, Shell & Scripting for FDEs

Assessment: Python Engineering Challenge

Backend Engineering, Observability & Advanced Data Systems

Introduction to Backend Engineering — HTTP, REST & API Design

FastAPI: Intro to Deep Dive

Auth: OAuth2, JWT, OIDC & RBAC

Building an Order Management Module & its API

Building Payments & Notifications (Stripe, Celery)

Building the Demo Frontend

Real-Time, Queues & Caching (WebSockets, SSE, Redis)

Observability & Distributed Tracing (OpenTelemetry, Grafana)

Postgres Advanced: JSONB, Full-Text Search & pgvector

Postgres Internals: EXPLAIN ANALYZE, Indexing & MVCC

NoSQL, Document DBs & Vector Databases

Dimensional Modeling, dbt & Modern Data Stack

Workflow Orchestration with Airflow & Data Quality

Event-Driven Architecture & Kafka Fundamentals

Assessment: Data Pipeline Build & Debug

Full-Stack FDE: TypeScript, React & AI-First Frontends

TypeScript Fundamentals for Python Engineers

React Essentials for AI Apps

Next.js for Full-Stack AI Apps

Vercel AI SDK & Streaming UIs

AI Product UX & Customer-Facing Polish

Cloud, DevOps, Kubernetes & Infrastructure

AWS Core: IAM, VPC & Networking

Compute, Storage & Managed Services (EC2, S3, RDS, SQS)

AWS Security & Secrets Management (KMS, Secrets Manager, CloudTrail)

Docker: Production Containers

Kubernetes Architecture & Core Concepts

EKS, Helm & Production Kubernetes

Kubernetes Networking, Security & GPU Workloads

CI/CD Pipelines & GitOps (GitHub Actions, ArgoCD)

Terraform & Infrastructure as Code

Serverless & Event-Driven Cloud (Lambda, API Gateway, Step Functions)

SRE & Reliability Engineering for FDEs

Assessment: Deploy, Break, Fix on Cloud

Enterprise Communication, Consulting & LLM Engineering

Customer Discovery & Requirements Gathering

Technical Scoping & Solution Documentation (PRD-lite, options matrix, risk register)
Stakeholder Communication & Written Artifacts
Demo Storytelling & Handling Pushback
LLM Fundamentals for Production
Prompt Engineering for Production Systems
RAG Architecture: Ingestion & Retrieval
RAG over Customer Documents & Hybrid Search
RAG Evaluation & Iteration (RAGAS)
LLM Safety, Guardrails & Security
Cost, Latency & Model Routing
Assessment: Discovery Call + RAG PoC Build

Agentic Systems & Enterprise Integrations

Tool Calling, Function Calling & Structured Output
Agentic Patterns & Production Concerns
Agent Debugging, Failure Analysis & Evaluation
MCP & Enterprise Tooling
Multi-Agent Patterns & Human-in-the-Loop
Assessment: Agentic System Build & Red-Team
Enterprise Systems Landscape & Integration Topology
API Integration Patterns & Connector Design
Integration Failure Modes & Debugging
Messy Customer Data & Entity Resolution
Build: CRM-to-AI Support Copilot
Assessment: Integration War Room

Application Engineering, Security & Reliability

Product Thinking & Architecture for FDE Apps
Frontend Essentials for FDEs: React + TypeScript
Multi-Tenant Design & Background Workflows (Postgres RLS, Celery)
Document Processing & AI Extraction Workflows
Build: Enterprise Document Review App
Developer Experience & API Documentation
Enterprise Security for FDE Systems (PII, SOC2, GDPR)
Secrets, Encryption & Deployment Topologies
Load Testing, Capacity Planning & Performance (k6)

Incident Response & Customer-Facing RCA

Enterprise Readiness Checklist & Go-Live

Assessment: Secure Deploy + Incident Simulation

System Design, Capstone & Hiring Readiness

Distributed Systems Fundamentals (CAP, PACELC, Raft, Saga)

System Design: RAG System at Scale

System Design: Agentic Platform at Scale

System Design: Multi-Tenant SaaS Platform

System Design: Enterprise Integration Hub

System Design Interview Framework & Practice

Capstone Sprint 1: Discovery & Solution Design

Capstone Sprint 2: Core Build & Integration Layer

Capstone Sprint 3: Harden, Secure & Observe

Capstone Sprint 4: Red-Team, Eval & Fix

Capstone Demo Day: Client Panel Presentation

Interview Readiness & Career Launch

FDE Mock Engagement: Full Simulation

Graduation Review: Portfolio Defense & Next Steps

Elective Containerization and Orchestration

Docker introduction

Docker image and Docker containers

Building your own Docker image

Image and Containers Deep Dive

Docker volume

Docker networking deep dive

Hardening the Docker env. (DevSecOps)

Demo of implementing industry standard implementation (e.g. in image hardening and other security standards)

Introduction to Orchestration

Kubernetes architecture

Kubernetes Pod deep dive

Kubernetes Replicaset and Deployment deep dive, side car

Kubernetes networking

Scaling options in kubernetes - HPA, CA and VPA - compare with Karpenter

Introduction to scaling in kubernetes, Network policies

Kubernetes security - RBAC (DevSecOps)

Kubernetes volumes

Elective AWS : multi-region cloud deep dive

AWS VPC & Networking Basics

Advanced AWS Networking

AWS Observability & CloudWatch

CloudWatch & CloudTrail Advanced (DevSecOps)

Docker App on EC2 Project

AWS Lambda & API Gateway

End-to-End Project day

AWS ECS & ECR Intro

ECS Fundamental

EKS Basics

EKS Advanced

EKS Security & Operations

Microservices on AWS Project

Microservices Project – Phase 2

AWS Security: Identity & Network (DevSecOps)

AWS Security: Data & Detection (DevSecOps)

AWS Security Hands-on (DevSecOps)

Elective Storage, Networking & Reliability for FDEs- storage architecture, DR, sharding, CDN, HA, fault tolerance

Architect the Storage for an Application

Data Lifecycle Management and Storage Optimization

Backup and Disaster Recovery Strategies

Designing the Networking of an Application

Load Balancing and Auto Scaling Groups

Designing for scalability of application and data

Database Sharding Strategies and Hot-Spot Mitigation

Edge Caching, CDNs, and Latency Reduction

Microservices Performance — Latency, Throughput, Concurrency

Designing for High Availability — Multi-Region Active-Active

Fault Tolerance Engineering — Circuit Breakers, Retries, Backoff

Graceful Degradation — Rate Limiting, Throttling, Shedding

System Design ●

Distributed System Design & AI-Integrated Architectures

System Design Foundations & Modern Infrastructure
Load Balancing, Consistent Hashing & Traffic Routing at Scale
Caching Systems: CDN, Backend Caches, Cache Invalidation
Case Study: Caching at Scale (Scaler Code Judge and Contest Leaderboards + AI Workloads)
Case Study: Caching Facebook News Feed
CAP / PACELC theorem + Master Slave Replication
SQL vs NoSQL + Sharding
Database Orchestration & Shard Creation
Case study: Google Typeahead (How to approach System Design Problems)
Case Study: Google Typeahead (Design and Optimizations)
NoSQL Internals - LSM Tree and Multi Master
Case Study: Messaging Apps (FB Messenger, Whatsapp, Slack)
Messaging Queues - Apache Kafka & Zookeeper
Case Study: Elasticsearch (Full Text Search)
Case Study: S3, HDFS (Large File Storage)
Case Study: Uber (Nearest Neighbor Search)
Case Study: Rate Limiter + Unique ID Generator (Infra)
Case Study: OTT Platform (OTT)
Microservices - 1
Microservices - 2
Case Study: Ecommerce Platform (Microservices)

Interview Preparation ●

Mastering LastMile - Interview Simulation

DSA Interview Simulation: Problem Solving & Thinking Under Pressure
LLD Interview Simulation: Structuring Scalable Solutions
HLD Interview Simulation: Designing Systems & Communicating Trad

Data Engineering ●

Data Engineering

Introduction to Data Engineering	
Writing Efficient Queries: Part 1	
Writing Efficient Queries: Part 2	
Intro to Hadoop	
HDFS and MAP-REDUCE	
Introduction to Hive	
Advanced features of Hive	
Basics of Spark	
Spark Programming	
Structured API - PySpark	
PySpark Operations	
Introduction to Streaming	
Apache Spark Streaming	
Advanced Spark Streaming	
Additional Streaming Services: Bringing it all together	
Modern Data Architectures	
AWS Services for Data Engineering	
Data Modelling	
Orchestrating data pipelines	
Orchestration Continued	
Distributed Coordination	
Load Balancing	
Data Pillars	
Capstone	

Advanced DSA ●

DSA Edge: Advanced Patterns & Interview Techniques

DSA Edge : Townhall	
DSA Edge : Townhall (Post-SQL - change name after scheduling)	
Revision of DSA Topics	
Maths: Inverse Mod & Problems	
Backtracking: Famous Problems	

Tries 1: Trie of Character

Tries 2: Trie of Bits + Problems on Trees

Strings Pattern Matching

DP 1: DP on Strings

DP 2: Famous Problems

Graphs 1: Bipartite Graph

Graphs 2: Belleman Ford & Floyd Warshall Algorithm

Advanced Interview Problems

DSA Edge Mandatory Contest: Full Syllabus

DSA Edge Full Syllabus: Contest Discussion

Product Management for Engineers

Townhall: Product Management

Introduction to Product Management 1

Understanding the Product Lifecycle 1

Understanding the Product Lifecycle 2

Critical Thinking in Product Development 1

Critical Thinking in Product Development 2

Market Structure Analysis and Its Technical Aspects 1

Market Structure Analysis and Its Technical Aspects 2

MVPs, Prototyping, and Growth Hacking 1

MVPs, Prototyping, and Growth Hacking 2

Concept Development, Validation, and the Role of Developers 1

Concept Development, Validation, and the Role of Developers 2

Introduction to Product Analytics 1

Gen AI for Software Engineers

Introduction to AI Engineering

Refresher: Modern Text & Image AI Model Architectures

Getting started with Text Generation LLM APIs (OpenAI API)

Designing Eval Pipelines for AI Apps and Observability

Prompt Engineering: Introduction

Retrieval-Augmented Generation (RAG) Introduction

Embeddings Deep Dive

Multimodal & Tabular RAG

Prompt Engineering: Security

Agents: Foundations & Planning

Advanced Agent Concepts

Agent: Frameworks and Protocols

Fine Tuning Existing Models

Audio, Image, Video, Expressions: AI Modalities

Scaling AI Applications

Model Quantization Techniques

Advanced Fine Tuning Techniques

Dataset Engineering & Inference Optimization